

AKSHAR VANDARA

Boston, MA | (857) 919-6234 | axr230102@gmail.com | [linkedin.com/in/akshar-vandara](https://www.linkedin.com/in/akshar-vandara) | github.com/akshar-23 | [portfolio](#)

EDUCATION

Northeastern University

Master of Science, Game Science and Design (GPA: 3.87 / 4.0)

May '26

Boston, MA

Gujarat Technological University

Bachelor of Engineering, Computer Engineering (GPA: 3.6 / 4.0)

May '23

Ahmedabad, India

EXPERIENCE

Research Assistant

Game Design Lab, Northeastern University

Feb '25 – Present

Boston, MA

- Modified the Sturgeon solver to generate 2D puzzle levels alongside valid solution paths.
- Optimized the generation algorithm to achieve 100x better performance and higher stability than prior academic methods.
- Co-authored a research paper detailing the system's architecture and performance metrics, published at the AAAI 2025 AIIDE conference. <https://doi.org/10.1609/aiide.v21i1.36818>
- Currently developing methods to generate 3D levels by blending procedural patterns extracted from multiple 2D games data.

Gameplay Systems & UI Programmer

Otisco Studios

May '25 – Aug '25

Orlando, FL

- Built 8+ core systems, including save/load and UI, across two distinct titles to support main game loops.
- Created 5+ NPC and enemy archetypes for roguelike and open-world genres using Behavior Trees and Blackboards.
- Resolved over 30 critical bugs during the pre-launch phase to ensure product stability for final release.
- Refactored codebases across multiple projects, resulting in a 100% increase in average gameplay framerates.

Unity Developer

Arcadon Games

Mar '23 – Jul '24

Bangalore, India

- Developed core mechanics for Cricket Tycoon using state machines, improving gameplay responsiveness by 30%.
- Integrated Firebase to manage 1,000+ user profiles and synchronize inventory data with real-time cloud updates.
- Designed simulation systems capable of processing 30 matches per minute, reducing total load times by 40%.
- Managed alpha testing for 100+ users, achieving 85% satisfaction and fixing 20+ critical stability issues.

PROJECTS

Spacetime Level Generation | [GitHub](#) | | [Publication](#) | | *Python, SAT Solvers, Pandas*

- Leveraged Python to optimize the 'Sturgeon' constraint-solver, integrating solution-path validation to automate the creation of playable 2D puzzles.
- Increased level generation speed by 100-500 times and improved error-proofing compared to previous methods like STWFC (Space-Time WaveFunctionCollapse)
- Co-authored and published a research paper on the system that was accepted into the 2025 AAAI conference (AIIDE 25).

Red Circuit Rebellion | [Steam](#) | | *Unreal Engine 5, C++, Blueprints, Behavior Trees*

- Developed 5+ core gameplay systems, including save/load, environmental fracturing, health management, and objectives manager.
- Utilized Unreal's Environment Query System (EQS) along with Behavior Trees to script tactical positioning and combat logic for 3 distinct enemy archetypes.
- Scripted 2 game modes (Story and Arcade) with specific objective logic and enemy spawners for each.
- Optimized game assets and blueprints, increasing average FPS by ~100% during gameplay.

Cricket Manager: League | [Playstore](#) | | *Unity, C#, Firebase, Firebase Realtime DB, Firebase Cloud Functions*

- Architected sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features.
- Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.

SKILLS

Game Dev:	Unity, Unreal Engine, Blueprints, Game Physics, Core-gameplay management, Database handling
Game Design:	Animations, Animation Rigging, UI integration, Environment Design, 3D Modeling
Languages:	C#, C++, JavaScript, Python, Java, R, C, SQL
Python Libs:	NumPy, Pandas, PyTorch, Cuda, TensorFlow, PyGames, Flask, Django, SQLAlchemy
Software Dev:	HTML, CSS, JavaScript, Three.js, Tailwind, React, Node, Vue, Angular, Next.js, Vite.js
Dev Tools:	Git, Perforce, Firebase Realtime DB, Firebase Cloud functions, SourceTree, Gitlab, Figma, Trello